

Woofers–tweeters temporal correction split in atmospheric adaptive optics

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Many adaptive optics applications require wavefront correctors with a high stroke, and at a high bandwidth. Often, these two requirements cannot be met by a single wavefront corrector, and, instead, the combination of a low-bandwidth, high-stroke woofer and a high-bandwidth low-stroke tweeter is used in a so-called woofer–tweeter architecture. The optimal (minimum residual phase variance) way to split the correction between the woofer and the tweeter in the context of a linear–quadratic–Gaussian (LQG) controller has been addressed previously. However, the necessity to fold the temporal characteristics of the woofer and tweeter into the LQG controller significantly increases its complexity. In this Letter, this optimal strategy is compared to a simpler, *ad hoc* approach, which consists in optimizing the LQG controller as if it were controlling a high-bandwidth, high-stroke corrector and splitting the correction using first-order high- and low-pass temporal filters. In the case of tilt correction for NFIRAOS on the Thirty Meter Telescope, it is found that the *ad hoc* approach, which is already used or planned for several systems, holds the same overall correction performance compared to the optimal strategy. © 2012 Optical Society of America

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Atmospheric turbulence is a major factor limiting the performance of ground-based telescopes [1]. It induces random fluctuations in the phase of light wavefronts propagating through the atmosphere, thus degrading image quality and reducing the resolving power of telescopes. An adaptive optics (AO) system can correct these undesirable effects by using a wavefront correction system (WCS) driven in real-time from the measurements of a wavefront sensor (WFS).

A WCS typically consists of a deformable mirror (DM), which is a reflecting flexible face sheet driven by a set of actuators underneath. Several types of DMs have been developed. They are characterized by their stroke (how much aberration they can correct) and their bandwidth (how fast they can correct it). It is often not possible to achieve both the required stroke and bandwidth in a single device, and most WCSs are relying on two correctors: one with a large stroke but a reduced bandwidth, called a woofer and one with a reduced stroke but a large bandwidth, called a tweeter. Such a woofer–tweeter scheme is particularly common for tip–tilt (TT) correction, because in atmospheric turbulence, TT (which induces image jitter) has a very large amplitude; besides, other effects intrinsic to the telescope, such as wind shake (WS), further add up. In this case, the woofer is a dedicated TT platform (TTS) holding a flat mirror (correcting only the low temporal frequencies of TT, for such a platform is usually large and heavy), and the tweeter is a deformable mirror (correcting the high temporal frequencies of TT, plus all the higher order terms). Such a woofer–tweeter architecture has been adopted in several state-of-the-art AO systems, including the Gemini Planet Imager [2] and NFIRAOS, the proposed first-light AO system for the Thirty Meter Telescope (TMT) [3].

So far, these systems have been relying on an *ad hoc* simple strategy to split the correction between the woofer and the tweeter: the AO temporal controller is designed as if there were only one high-stroke, high-bandwidth

corrector, and the split is performed independently of the AO controller by a pair of filters: a first-order low-pass filter for the commands to the woofer and a complementary high-pass filter for the commands to the tweeter—in other words, the input–output transfer function of this dual corrector is assumed to be equal to 1, a convenient simplification for the AO design.

Recently, the Strehl-optimal minimum-variance (MV) solution to this problem has been proposed [4]. It is based on a detailed and accurate knowledge of the temporal dynamics of the AO loop, as well as the spatial statistics of the disturbance and measurement noise. It builds upon the linear–quadratic–Gaussian approach and is designed in state-space instead of the frequency domain.

This optimal approach, however, results in a more complex controller. It would therefore be interesting to compare its performance for TT modes to the simple *ad hoc* suboptimal solution that has been used so far, to understand if the additional complexity could yield meaningful improvements.

This work was initially motivated by the correction of image jitter of NFIRAOS, Thirty Meter Telescope's (TMT) first-light laser-guide-star-based multiconjugate AO facility, [3]. Though parameters are tailored for NFIRAOS, the conclusions remain quite general. Table 1 summarizes the parameters used in the detailed Monte Carlo simulations.

The atmospheric TT and WS values are typical of the conditions expected at TMT on Mauna-Kea. The photon detection equivalent (PDE) parameter covers a broad range of natural guide stars (NGS) used for sensing TT in a NFIRAOS-like configuration, from magnitudes 12 to 22 with a telescope throughput of 0.4 and 10% Strehl at the edge of the field.

The system uses a single TT sensor. The AO controller is optimized based on the WFS flux levels, which sets the signal-to-noise ratio on the WFS, including the frame rate in the range of 20 to 800 Hz. Gaussian noise added to

Table 1. Parameters Used in the End-to-End Monte Carlo Numerical Simulations

Disturbances	
Atmospheric	$\sigma_{TT} = 16.58$ [mas rms] $L_0 = 30$ m $r_0@0.5 \mu\text{m} = 0.15$ m
Wind shake	$\sigma_{WS} = 7.5$ [mas rms]
WFS	
Frame rates	$\{20 \dots 800\}$ [Hz]
Photon detection equivalent	$\{5 \times 10^2, \dots, 5 \times 10^6\}$ [e^-/s]
Readout noise	$\{3.1 - 7.4\}$ [e^-]
Wavelength	$\lambda = 1.6 \mu\text{m}$
Correctors	
TT mount	90 Hz bandwidth (-3 dB)
DM	infinite bandwidth
Simulation	
Duration	5×100 [s]
Suboptimal split 1	LP/HP with 20 Hz cutoff
Optimal split 2	global MV controller

the measurements follows formulas in [5] for a quad-cell detector in the H band.

The NFIRAOS-customized TTS will support a 60×60 actuator deformable mirror. This TTS is characterized by a second-order dynamical model with an undamped natural frequency of $f_0 = 82$ Hz and a damping ratio

of $\xi = 0.48$. The NFIRAOS DM (tweeter) has several kilohertz of bandwidth and can be considered infinitely fast.

The two configurations compared in this study are depicted in Fig. 1. The top panel shows the block diagram using a hybrid discrete- and continuous-time implementation. The bottom panel shows the *ad hoc* temporal-frequency split with the split frequency fixed to $f_c = 20$ Hz and the MV controller addressing both the TTS and the DM altogether. Further details on the design of the MV controller can be found in [4].

Figures 2(a) and 2(b) clearly show that both configurations achieve the same performance [expressed in residual TT error in milliarcseconds (mas) rms] at all flux levels. A slight difference in favor of the LP/HP solution is found at a high PDE level, though such differences are in the order of the hundreds of microarcseconds, a manifestation of the pure delay of 1 ms not accounted for in the optimal solution. The rms relative error obtained between both solutions is below 1% for all scenarios tested. Overplotted in Figs. 2(a) and 2(b) are the minima achieved by both configurations as well as the temporal frame rate used to obtain them. In Fig. 2(a) the equivalent star magnitude in the H band is also plotted for reference, using $\text{PDE}(m_v) = 30^2 \pi / 4 \times 3.17 \times 10^9 \times \chi \times 10^{-(k+m_v)/2.5}$, with $\chi = 0.4$ being the telescope throughput and $k = -\log(0.98)$.

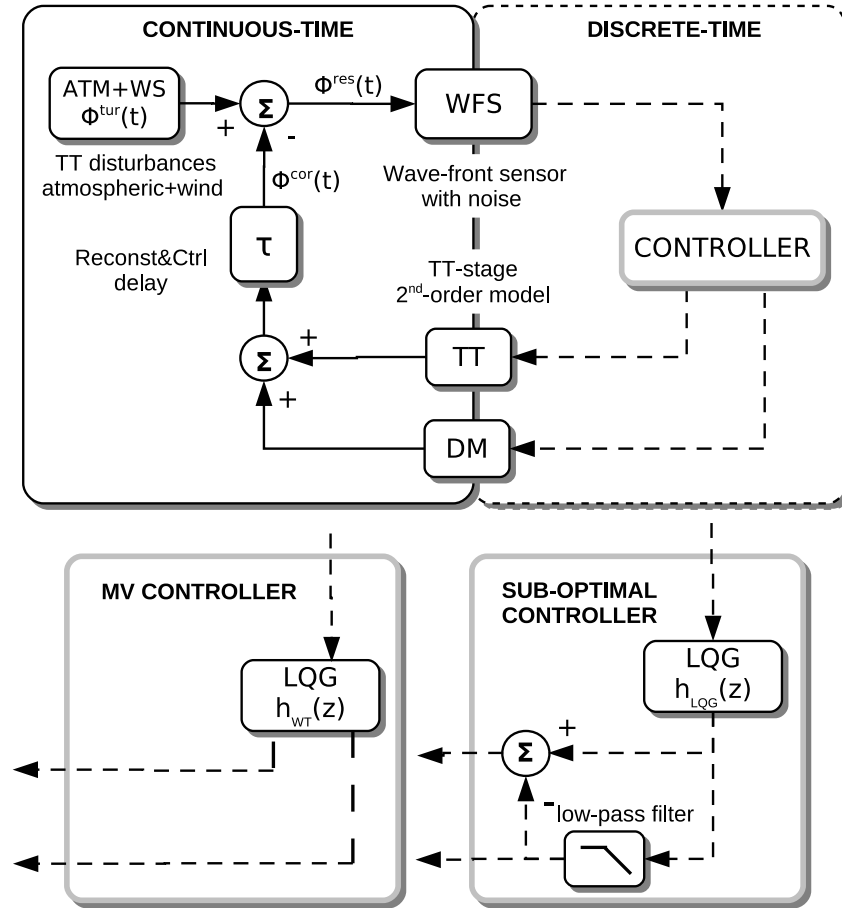


Fig. 1. Block diagram of a hybrid continuous-discrete model with continuous uncorrected disturbances and phase correction (solid lines) and discrete measurements and commands (dashed lines). The bottom-left panel depicts the MV controller, and the bottom-right panel shows the suboptimal *ad hoc* solution using first-order single-pole low-pass/high-pass split correction.

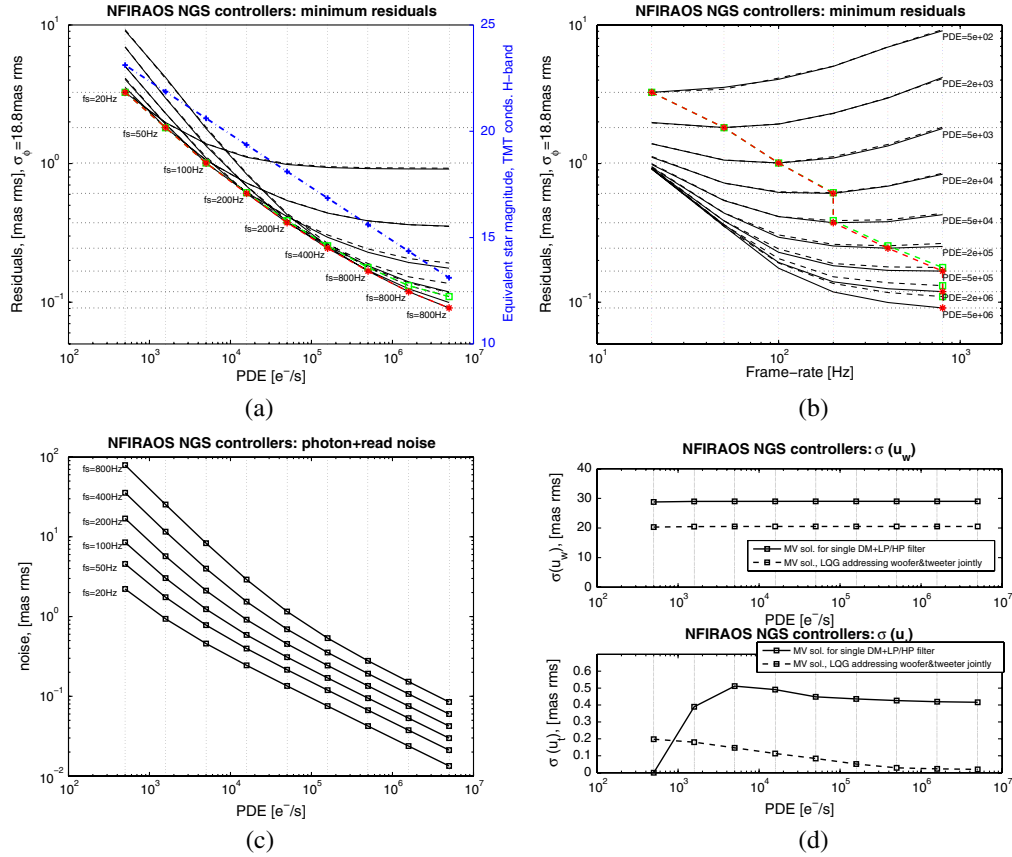


Fig. 2. (Color online) Top panel, (a) Overall performance in mas rms (b) Optimal frame rates. Solid lines show the MV solution with a single DM + LP/HP filter and dashed lines show the MV solution addressing the woofer and tweeter jointly. The minima and the frame rates with which they were obtained are also indicated in red and green, respectively. Bottom panel, (c) photon + read noise added to the measurements. (d) Effort in mas rms of the correction demanded from the TTS and the DM.

The optimal MV solution has more flexibility though. The bottom panel of Fig. 2 shows the amount of correction that is assigned to each of the woofer and the tweeter. In the MV solution [4], one can assign to the tweeter a predetermined maximum fraction of the correction to be applied to the woofer, which can be a very convenient feature. In the example, 10% of the rms value was chosen; it is reached only for very low magnitudes (or equivalently low PDE or low signal-to-noise ratio, on the left of the plots). In the *ad hoc* solution, the amount of correction assigned to the tweeter can be changed by adjusting the cutoff frequency of the LP/HP filters. However, if too high a cutoff frequency is used to reduce the demand on the tweeter, the woofer can become limited by its own temporal bandwidth, as is shown in [6] in the context of the Gemini Planet Imager. Finally, note that the MV solution requires significantly less of a stroke on both the woofer and the tweeter. Although this is usually not significant, it can become important if the stroke of the woofer or of the tweeter are very limited.

In conclusion, the Strehl-optimal correction with woofer-tweeter systems in atmospheric AO can be approximated, with unnoticeable performance difference by using a simpler temporal *ad hoc* solution. This simplification consists in optimizing the minimum-variance LQG controller as if it were controlling a high-bandwidth, high-stroke corrector, and splitting the correction using

simple high- and low-pass temporal filters with a cutoff frequency chosen relatively below the woofer's -3 dB temporal bandwidth. It does, however, require a higher stroke from the woofer and is less flexible than the MV solution addressing the woofer and tweeter jointly in that only the temporal filter cutoff frequency can be adjusted to balance the amount of offload.

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